EM Algorithm – 1-d EM in Python

In this exercise, you will be implementing the EM Algorithm for the 1-dimensional case.

a) Start your dev environment and load the Jupyter Notebook for this Exercise.

b) Read and understand the given code.
- How many and which data point do you have?
- How many clusters are you supposed to find?

c) Complete the `TODO`s in the code. Your initialization for the clusters should give them equal weight, an equal variance of 1 and starting means of 0 and 1 respectively. Run and debug your implementation. Here is the result you should be getting:

```
Iteration  0, L= -132.779820:
    r[ 0][ 0]=0.000000  r[ 1][ 0]=0.000000  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.000000  r[ 1][ 1]=0.000000  r[ 2][ 1]=0.000000  r[ 3][ 1]=0.000000  r[ 4][ 1]=0.000000
    Cluster  0 with mean=0.000000 variance=1.000000 weight=0.500000
    Cluster  1 with mean=1.000000 variance=1.000000 weight=0.500000

Iteration  1, L= -13.058782:
    r[ 0][ 0]=0.182426  r[ 1][ 0]=0.075858  r[ 2][ 0]=0.000203  r[ 3][ 0]=0.000075  r[ 4][ 0]=0.000028
    r[ 0][ 1]=0.817574  r[ 1][ 1]=0.924142  r[ 2][ 1]=0.999797  r[ 3][ 1]=0.999925  r[ 4][ 1]=0.999972
    Cluster  0 with mean=2.302134 variance=0.267765 weight=0.051718
    Cluster  1 with mean=7.256215 variance=13.479628 weight=0.948282

Iteration  2, L= -12.282043:
    r[ 0][ 0]=0.476244  r[ 1][ 0]=0.233815  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.523756  r[ 1][ 1]=0.766185  r[ 2][ 1]=1.000000  r[ 3][ 1]=1.000000  r[ 4][ 1]=1.000000
    Cluster  0 with mean=2.329290 variance=0.220858 weight=0.142012
    Cluster  1 with mean=7.773083 variance=12.072180 weight=0.857988

Iteration  3, L= -11.299169:
    r[ 0][ 0]=0.791962  r[ 1][ 0]=0.531722  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.208038  r[ 1][ 1]=0.468278  r[ 2][ 1]=1.000000  r[ 3][ 1]=1.000000  r[ 4][ 1]=1.000000
    Cluster  0 with mean=2.401698 variance=0.240337 weight=0.264737
    Cluster  1 with mean=8.655652 variance=8.599896 weight=0.735263

Iteration  4, L= -9.806763:
    r[ 0][ 0]=0.952890  r[ 1][ 0]=0.867866  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.047110  r[ 1][ 1]=0.132134  r[ 2][ 1]=1.000000  r[ 3][ 1]=1.000000  r[ 4][ 1]=1.000000
    Cluster  0 with mean=2.476651 variance=0.249455 weight=0.364151
    Cluster  1 with mean=9.590527 variance=3.446268 weight=0.635849

Iteration  5, L= -8.466513:
    r[ 0][ 0]=0.999827  r[ 1][ 0]=0.998511  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.000173  r[ 1][ 1]=0.001489  r[ 2][ 1]=1.000000  r[ 3][ 1]=1.000000  r[ 4][ 1]=1.000000
    Cluster  0 with mean=2.499671 variance=0.250000 weight=0.399668
    Cluster  1 with mean=9.996066 variance=0.694285 weight=0.600332

Iteration  6, L= -8.465259:
    r[ 0][ 0]=1.000000  r[ 1][ 0]=1.000000  r[ 2][ 0]=0.000000  r[ 3][ 0]=0.000000  r[ 4][ 0]=0.000000
    r[ 0][ 1]=0.000000  r[ 1][ 1]=0.000000  r[ 2][ 1]=1.000000  r[ 3][ 1]=1.000000  r[ 4][ 1]=1.000000
    Cluster  0 with mean=2.500000 variance=0.250000 weight=0.400000
    Cluster  1 with mean=10.000000 variance=0.666667 weight=0.600000
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Needed 6  Iterations
    Cluster  0 with mean=2.500000 variance=0.250000 weight=0.400000
    Cluster  1 with mean=10.000000 variance=0.666667 weight=0.600000
```